# Predicting Acute Myeloid Leukemia (AML) Status from HLA Genotypes Using Classical Machine Learning

Pouya Mirzaeizadeh Shadi Sefidgar Dr. Kherad Pishe

November 25, 2025

**Abstract**

This study investigates whether human leukocyte antigen (HLA) genotypes, combined with demographic information (age and sex), can predict acute myeloid leukemia (AML) status using classical machine learning models. Despite extensive preprocessing, encoding, and model comparison, the results show that HLA-based prediction of AML is not feasible with the current dataset. Models achieved superficially high accuracy by predicting the majority class (AML) but failed to detect healthy individuals, yielding near-zero recall for the minority class. The findings highlight methodological challenges in tabular ML under severe class imbalance and limited sample size, offering insights for future computational immunogenetics research.

# 1 Introduction

Acute Myeloid Leukemia (AML) is a heterogeneous hematologic malignancy with complex genetic and immunologic underpinnings. This project explores whether HLA genotypes—alongside age and sex—can serve as predictive features for AML status using classical machine learning (ML) techniques. The study emphasizes methodological rigor and error analysis over predictive success, revealing the limitations of tabular ML in imbalanced biomedical datasets.

# 2 Dataset and Variables

## 2.1 Source and Structure

The dataset comprised 294 individuals with 16 usable columns after cleaning. Each row included:

- Age (numeric)
- Sex (binary: male = 1, female = 0)
- Disease status (binary: AML = 1, healthy = 0)
- HLA loci: A1, A2, B1, B2, C1, C2, DRB3, DRB4, DRB5, DQB11, DQB12

## 2.2 Class Imbalance

The dataset was heavily imbalanced, with AML cases forming the majority. This imbalance posed a major challenge for model generalization and minority class detection.

# 3 Preprocessing and Feature Engineering

## 3.1 Cleaning and Recoding

Empty columns and identifiers were removed. Labels were recoded for binary classification. All features were numeric after preprocessing.

## 3.2 Handling Missing Values

Two datasets were constructed:

- **Dataset A**: Complete-case HLA profiles (245 samples, 153 features)
- **Dataset B**: Reduced-locus dataset excluding C1, C2, DQB11, DQB12 (272 samples, 118 features)

## 3.3 Encoding

HLA alleles were one-hot encoded. Age and sex remained numeric. Stratified train-test splits (80/20) and 5-fold cross-validation were used.

# 4 Modeling Approach

## 4.1 Models Tested

We evaluated:

- Logistic Regression, Naive Bayes variants, SGDClassifier
- k-Nearest Neighbours
- Random Forest
- Gradient Boosting, LightGBM, XGBoost
- Support Vector Classifier

## 4.2 Hyperparameter Tuning

GridSearchCV was used to optimize parameters for Naive Bayes, Logistic Regression, Random Forest, and Gradient Boosting models.

## 5 Evaluation Metrics

Performance was assessed using:

- Accuracy
- Precision, Recall, F1-score (per class)
- Macro averages
- Confusion matrices

Recall for the healthy class was prioritized due to clinical relevance.

## 6 Results

### 6.1 Representative Performance

Table 1: Representative model performance

| Dataset | Model | Accuracy | Recall (Healthy) | Recall (AML) |
|---|---|---|---|---|
| A | Naive Bayes (test) | 0.88 | 0.00 | 1.00 |
| B | Naive Bayes (test) | 0.89 | 0.00 | 1.00 |
| A | Gradient Boosting (test) | 0.88 | 0.00 | 1.00 |
| All | Gradient Boosting (CV) | 0.84 | 0.11 | 0.94 |
| All | LightGBM (CV) | 0.85 | 0.10 | 0.95 |

### 6.2 Interpretation

Most models collapsed into predicting AML for all samples. Ensemble methods showed slight improvement in cross-validation but remained clinically inadequate.

## 7 Feature Importance and Interaction Analysis

### 7.1 Gradient Boosting Importance

Tree-based importance scores highlighted Age and a few HLA alleles. These results are exploratory due to poor recall.

### 7.2 SHAP Analysis

SHAP values confirmed that Age and certain alleles pushed predictions toward AML, reflecting model bias.

### 7.3 Permutation Importance

Permutation importance revealed B2_40.0, DRB5, and DQB12_3.0 as top contributors. However, their predictive utility remains limited due to imbalance.

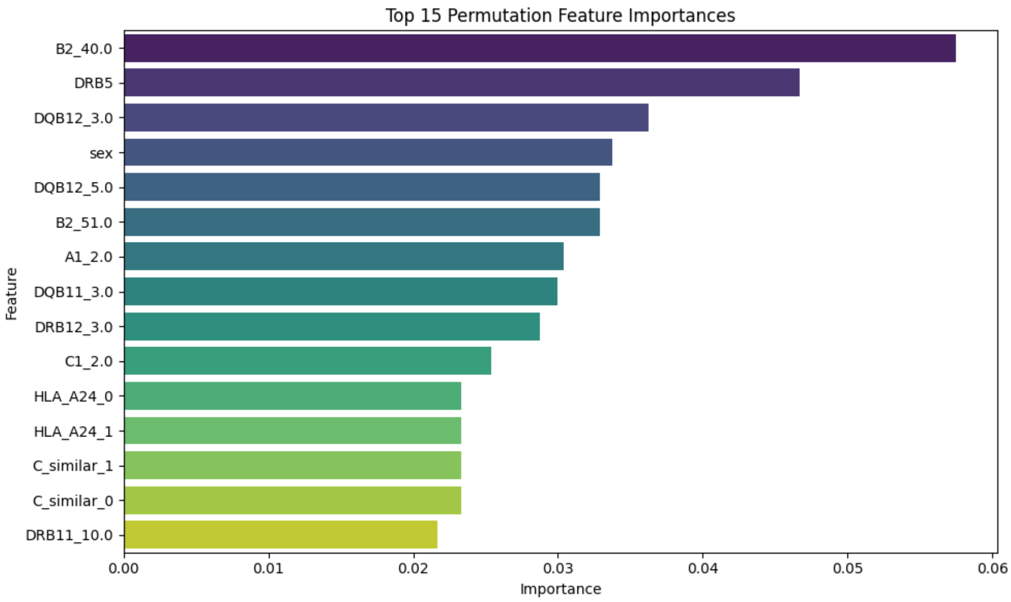


Figure 1: Top 15 Permutation Feature Importances

### 7.4 Feature Interaction Heatmap

The heatmap shows strong interaction between A1 and A2, and moderate interactions among B1, B2, and DRB loci. These patterns may inform future feature engineering.

### 7.5 Additional Feature Selection

We applied:

- Mutual Information
- Recursive Feature Elimination
- Boruta
- Random Forest impurity-based importance

Age and select HLA alleles consistently ranked high, but none yielded robust classifiers.

## 8 Error Analysis

Misclassified samples were scattered across age and allele distributions. No coherent borderline group emerged, reinforcing the absence of a learnable signal.

## 9 Discussion

### 9.1 Why Models Failed

1. **Class imbalance**: Majority-class dominance led to trivial classifiers.
2. **Sample size**: Sparse high-dimensional features hindered generalization.
3. **Label/genotyping noise**: Potential errors further diluted signal.

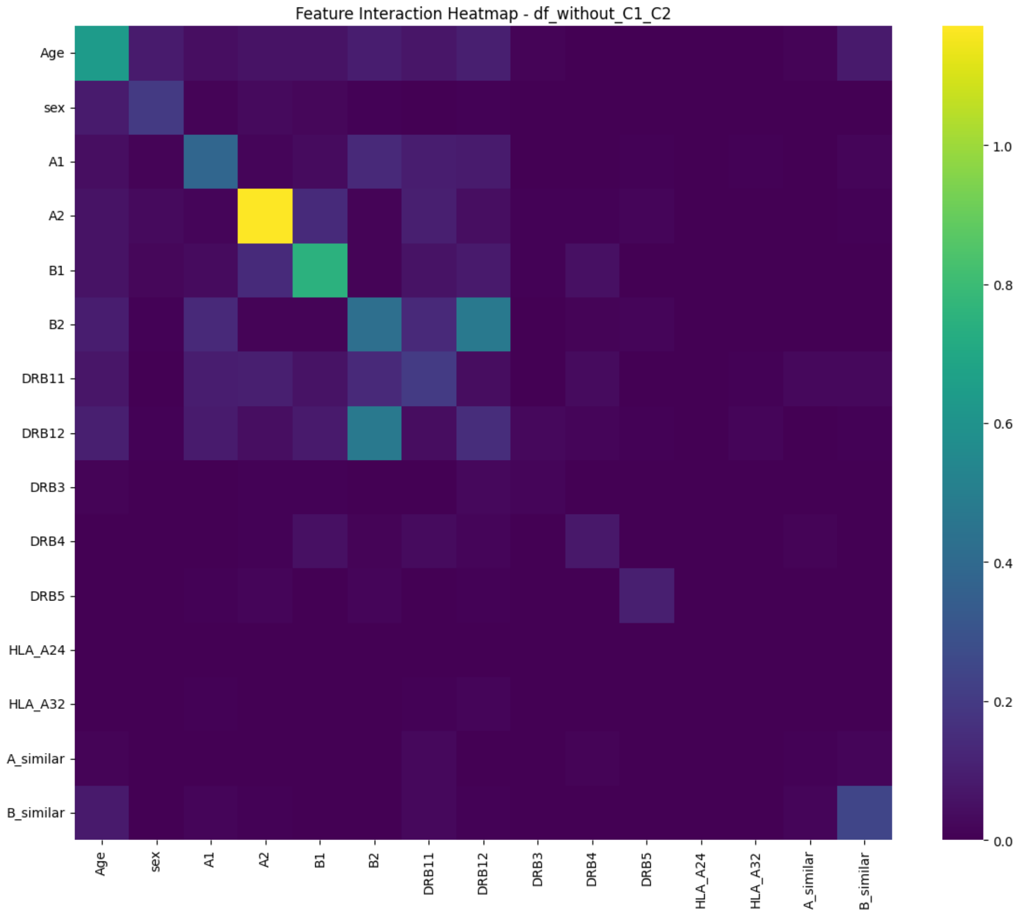


Figure 2: Feature Interaction Heatmap – Dataset B (without C1, C2)

## 9.2 Methodological Insights

This study underscores the need for:

- Advanced imbalance-handling (e.g., SMOTE, cost-sensitive learning)
- Dimensionality reduction or allele grouping
- Larger, more balanced cohorts
- Integration of clinical and genomic features

# 10 Conclusion

Despite rigorous preprocessing and modeling, HLA genotypes alone do not yield reliable AML classification using classical ML. The project highlights the limitations of tabular ML under imbalance and sparsity, and offers a foundation for future work in computational immunogenetics.

# References

[1] Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems.

[2] Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Advances in Neural Information Processing Systems.